CORL 2026 WORKSHOP · HALF DAY

CoRL 26
Conference on Robot Learning

From GPU-Accelerated Simulation to Scalable and Generalizable Real Robot Policy Learning

November 12, 2026 • Austin, Texas, USA • Half-Day Workshop

• https://corl2026-realworld-robot-learning.github.io

> *"Simulation is the most scalable training ground for embodied AI — from rigorous benchmarks and diagnostic evaluation protocols, to reliable transfer to real-world deployment."*

WORKSHOP OVERVIEW

SCAN FOR FULL PROGRAM

Simulation has become the engine of modern embodied AI. Physics-grounded environments like **BEHAVIOR** deliver the scale, safety and reproducibility that physical data collection cannot. Diagnostic frameworks like **RoboEval** instrument every task with stage-level metrics that reveal not just whether a policy succeeds, but precisely how and why it fails. **RoboChess** puts simulation-trained policies to the ultimate test by requiring direct transfer to physical robots. The frontier question is not whether to train in simulation, but **how to ensure that capabilities learned at scale carry over robustly to the physical world.**

CALL FOR PAPERS — TOPICS OF INTEREST

- Simulation environments and benchmarks for scalable robot learning and evaluation
- Diagnostic and fine-grained evaluation: capability metrics, failure analysis, robustness, reproducibility
- Sim-to-real transfer and validation: domain shift, adaptation, simulated-vs-real performance studies
- Challenging robot capabilities: long-horizon, dexterous, bimanual and generalist robot learning

INVITED SPEAKERS

Yashraj Narang
NVIDIA

Guannan Qu
Carnegie Mellon University

Yilun Du
Harvard University

Chen Tang
UC Los Angeles

KEY DATES

Submission Open	Aug 26, 2026
Paper Deadline	Oct 9, 2026
Notification	Oct 23, 2026
Workshop Day	Nov 12, 2026

Embodied AI Benchmark & Sim-to-Real Challenge Suite

Three tracks · independent leaderboards · shared evaluation protocol

TRACK 1 BEHAVIOR
Long-Horizon Household Challenge

1,000 household activities across 50 scenes and 10,000 objects on the Isaac Sim–based BEHAVIOR simulator — reasoning, navigation and dexterous bimanual manipulation over long horizons.

behavior.stanford.edu

TRACK 2 RoboEval
Diagnostic Bimanual Challenge

Stage-level instrumentation beyond binary success — outcome, efficiency, bimanual coordination and safety. 3,000+ VR-teleoperated expert demonstrations.

robo-eval.github.io

TRACK 3 RoboChess
Sim-to-Real Manipulation Challenge

Policies trained in GPU-accelerated simulation must transfer to physical hardware — perceive a real chessboard, plan long-horizon moves, execute across embodiments.

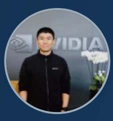
sites.google.com/view/corl-workshop-robochess

GENERAL ORGANIZERS

Zhenzhen Li
NVIDIA

Yizhou Zhao
NVIDIA

Jianwen Xie
Lambda



Zijian (Leo) Du
NVIDIA

Ruohan Zhang
Northwestern University

Huang (Raven) Huang
Stanford University

Jiafei Duan
NUS

Jiajun Wu
Stanford University

Call for Papers

- **Format:** short papers, up to 5 pages excluding references
- **Review:** double-blind on OpenReview
- **Presentation:** spotlights & posters, interactive discussion
- **Welcome:** methods, benchmarks, challenge reports, empirical studies, negative results, position papers

SUBMIT ON OPENREVIEW

CoRL 2026 · SimBench2Real

CHALLENGE ORGANIZERS

TRACK 1 · BEHAVIOR Fei-Fei Li · Jiajun Wu · Ruohan Zhang · Huang Huang · Wensi Ai · Stef Ren · Cem Gokmen · Yalcin Tur · Minyeong Kim · Andi Xu · Lynn Jin · Brenda Chen · Sanjana Srivastava · Chengshu Li · Josiah Wong · Hang Yin

TRACK 2 · RoboEval Yi Ru Wang · Jiafei Duan · Manling Li · Ranjay Krishna · Dieter Fox · Siddhartha Srinivasa · Carter Ung · Christopher Tan · Grant Tannert · Josephine Li · Amy Le · Rishabh Oswal · Markus Grotz · Wilbert Pumacay · Yuquan Deng

TRACK 3 · RoboChess Zhenzhen Li · Yizhou Zhao · Jianwen Xie · Zijian (Leo) Du · Fangzhou Mu · Yuheng Li

CoRL 2026 Workshop · **SimBench2Real**

Contact **zhenzhenl@nvidia.com**

PARTICIPATING INSTITUTIONS

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